

JYOTSNA VENKATESAN

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Education

The Hong Kong Polytechnic University

BSc (Hons) in Computer Science

2022 – 2026

Full Tuition Scholarship

Honors & Awards

- Full Tuition Scholarship** — Merit-based scholarship covering full undergraduate tuition 2022 – 2026
- HKSAR Reaching Out Award** — Awarded by HKSAR for contributions to international student outreach 2024
- Cathay Pacific Hackathon Semi-finalist** — Semi-finalist among 1500+ participants 2024
- Micro Fund Shortlisted Team** — Admission to HKSTP's Ideation Program and received HK\$20K 2025
- Selected for PolyU Delegation** — Chosen to represent PolyU at the International Astronautical Congress 2025 in Sydney 2025

Skills

- Languages:** Python, Java, C++, C, JavaScript, TypeScript, Swift, HTML/CSS
- Web & Frontend:** React, Vue, Svelte, Tailwind CSS, Figma
- Backend & DevOps:** Node.js, MySQL, MongoDB, REST APIs, Docker, Git
- AI:** Large Language Models (LLMs), Model Context Protocol (MCP) servers, and APIs including Azure OpenAI, Claude, and GPT
- Testing & Tools:** Selenium, Appium, JUnit, TestNG, Bash, MIPS

Experience

AI Software Developer Intern

June 2025 – Present

Career Hackers

Hong Kong

- Developed features for an AI interview platform with TypeScript using the MERN stack.
- Designed data schemas and interfaces for the platform with Figma.
- Built a resume parser, a Model Context Protocol server, and implemented search techniques including BM25, HNSW, and RRF using ElasticSearch.

Co-founder and CTO

May 2025 – Present

Orbityr

Hong Kong

- Founded an AI-powered location intelligence company.
- Accepted into HKSTP's 1-year Ideation Programme (HKD 100,000 grant) and secured additional HKD 20,000 funding.

Part-time QA Intern

September – December 2024

GoGoX

Hong Kong

- Automated 90% of regression tests, reducing manual testing time from 2.5 days to 3 hours using Java and Appium.
- Improved test coverage through data-driven testing techniques.

Quality Assurance Intern

June – August 2024

GoGoX

Hong Kong

- Automated 100% of sanity test suite, reducing execution time by 25% with Java and Appium.
- Expanded automated test coverage across 3 platforms and 3 countries.
- Participated in scrum meetings, gained understanding of agile methodologies.

UI/UX Designer Intern

February 2024 – May 2024

PolyU Smart Cities Research Institute

Hong Kong

- Designed QC Spatial-3Dx, a 3D camera application, using Figma and Adobe Photoshop.
- Led front-end development using React and Three.js for a responsive website.

Web/Front-End Developer Intern

June 2023 – August 2023

Oxbridge Economics

Remote

- Designed and developed web apps using Figma and React.
- Implemented RESTful API connections, improving data flow efficiency between front-end and back-end.

Student Assistant

February 2023 – April 2023

CLP

Hong Kong

- Contributed to the development of PolyU's Generator Inspection Robot for CLP.
- Conducted testing of the robot at CLP's power plant.

Projects

Sporkify - Recipe Management Application

Vue.js, Node.js, SQLite3, Tailwind CSS, Axios, Express.js

- Developed a full-stack recipe management app with secure user authentication, CRUD operations, and responsive UI.
- Implemented advanced search, category-based filtering, and image uploads with RESTful API endpoints using Node.js and Express.
- Designed structured SQLite3 database with tables for users, recipes, and metadata for efficient querying.

Orbityr - Location Intelligence

SvelteKit, TypeScript, Node.js, Tailwind CSS, MCP

- Built an AI-powered location intelligence platform using SvelteKit and TypeScript for seamless front-end/backend integration.
- Implemented MCP-based infrastructure for efficient real-time data retrieval and AI model orchestration.
- Designed scalable full-stack architecture with Tailwind CSS for responsive, modern UI.

Cosmic Soundscape Generator

Python, MIDI, Computer Vision, NumPy, OpenCV, music21

- Generated MIDI music from images using computer vision techniques to map color, texture, and shapes to tempo, scale, instruments, and chord progressions.
- Created multi-track compositions with algorithmic harmony; output project referenced by Samsung Research India for integration and citation.

Secure Online Storage System

Python, AES-GCM, SQLite3, PBKDF2, TOTP, Flask

- Developed a client-server secure file storage system with end-to-end AES-GCM 256-bit encryption, two-factor authentication, and strict access controls.
- Implemented audit logging, parameterized queries, and robust defenses against SQL injection, path traversal, and unauthorized access.

ZZZAFARI - Sleep Tracker Mobile App

SwiftUI, Swift, Core Data, Xcode, Krea

- Developed a sleep tracker mobile app with an interactive animal reward system dynamically reflecting user sleep patterns.
- Created all custom animal assets and AI-assisted backgrounds using Krea to enhance UI/UX design.
- Implemented persistent storage with Core Data and interactive SwiftUI animations for engaging user experience.

Leadership & Activities

Member, Rewriting the Code

2025 – Present

President, PolyU Hall Community Service Group

2024 – 2025

Vice President, PolyU SAO International Student Association

2023 – 2024

Powertrain Member, PolyU E-formula Racing (Powertrain)

2023 – 2024

Technical Mapping Team Member, PolyU Robocon Team

2022 – 2023

Workshop Facilitator, ProjeKt Inspire, Tanzania

2024